

6. SECOND WEEK OF EMBRYONIC DEVELOPMENT

The second week of embryonic development is marked by two major events:

- Implantation (nidation) of the blastocyst in the endometrium.
- Pre-gastrulation or transformation of the blastocyst into a bilaminar embryo (2 layers), at the same time as the appearance of the extraembryonic membranes.

1. NORMAL LOCATION AREA

Normally, implantation takes place in the upper third or middle third of the uterine wall, on its posterior side, during the progestational (secretory) phase of the menstrual cycle.

There ectopic pregnancy (not in its usual place) of the fertilized egg is due to a delay in the migration of the egg.

(mobility problem, permeability problem, of organs).

Most often implanted in the Fallopian tube,

The zygote can end its race on a ovary, abdominal cavity or directly onto the cervix.

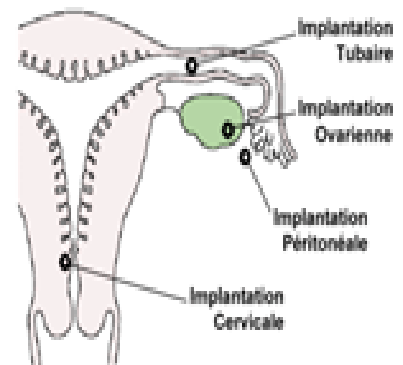


Figure 1. Normal implantation area

2. IMPLEMENTATION STAGES

In schematic terms, we can distinguish 3 stages:

2.1. Apposition of the blastocyst to the uterine mucosa

When the blastocyst is released from its zona pellucida on the 5th day, it comes into contact with the maternal uterine lining, facing the endometrium on the side of its pole embryonic. However, implantation can only occur if the uterus has entered its secretory phase. This phase of endometrial receptivity, which lasts 4 days (days 20-23), is commonly called the "implantation window".

It is characterized by the appearance of microprotrusions at the apical pole of the epithelial cells of the endometrium whose functional role would be to bring it closer to the blastocyst of the endometrium by absorbing uterine fluid.

2.2. Adhesion of the blastocyst to the endometrium

The microvilli on the surface of the outermost trophoblastic cells establish an interpenetration system with those of the uterine epithelial cells. Junctional complexes are

formed, resulting in stronger adhesion. The adhesion of the blastocyst to the endometrium is thought to be mediated by surface glycoproteins, the mechanism of which remains controversial.

2.3. Trophoblast invasion

As it progresses through the endometrium, the ST extends to the surface of the egg and will completely surround the CT.

Between the 9th and 12th days, the chorionic sphere penetrates completely into the endometrium and the ST layer then completely surrounds the CT layer.

and as it thickens, there appear gaps that communicate with the vessels of The endometrium on day 14 marks the l utero-lacunar circulation.

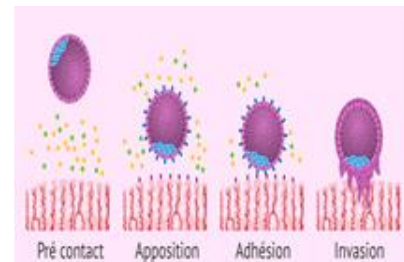


Figure 2. Implementation phases

3. ENDOMETRIAL RESPONSE

The uterine lining reacts to implantation by the decidual reaction which transforms the endometrium in deciduous stages.

The decidual reaction first occurs in the vicinity

From the embryonic pole: the modified cells form at this level the basal decidua which will participate later to the formation of the placenta. Then there are all

the connective tissue cells adjacent to the trophoblast which

are in turn modified and form the reflected or ovular decidua. Finally, the entire uterine endometrium is modified and forms the parietal decidua.

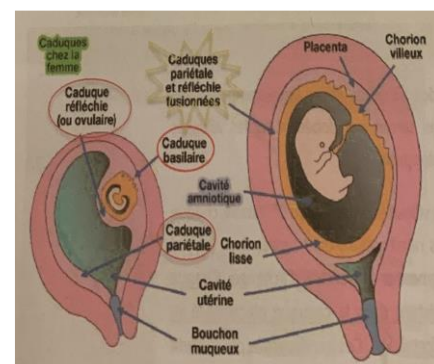


Figure 3. Formation of deciduous

4.PREGASTRULATION

4.1. Trainingof the two primary leaflets (epiblast and hypoblast)

On the 8th day, the superficial cells of the embryonic bud, in contact with the blastocoel, tend to flatten and form the first embryonic layer: the entophyll (hypoblast or primary endoderm).

The deep cells of the embryonic disc become taller and cylindrical to form the ectophyll (epiblast): ectophyll corresponds to the future dorsal side of the embryo and the entophyll is located on the side of the future ventral side of the embryo.

The whole thing has the shape of an oval disc, the bilaminar embryonic disc which will not change in appearance during the entire 2nd week of development.

4.2. Development of embryonic annexes

➤ Training of the amniotic cavity and the amnion

Ectophyll detaches by cleavage between cells. The intercellular space widens, revealing a fluid-filled cavity, the amniotic cavity. Flattened cells are born from the CT; these are the amnioblasts, which form a simple squamous epithelium that lines the roof of the amniotic cavity, the amnion.

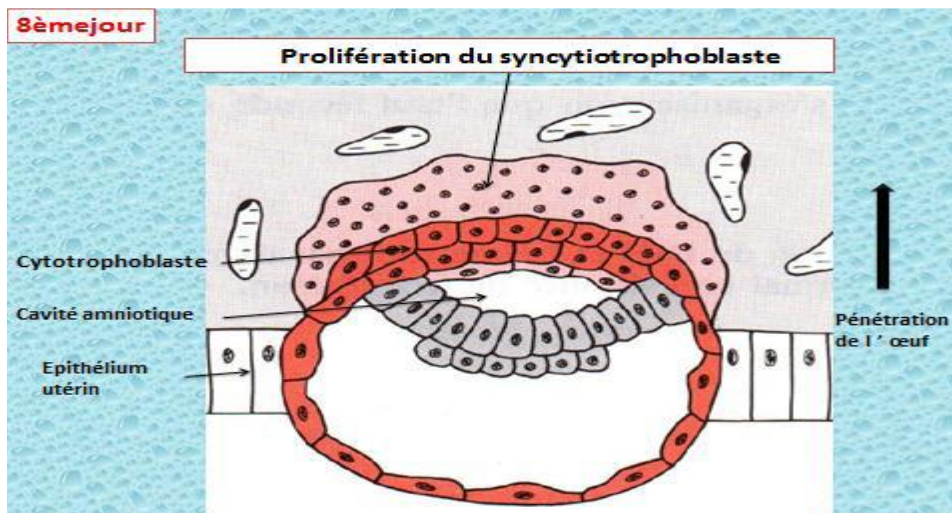


Figure 4. Formation of the amniotic cavity

➤ Formation of the primary yolk sac

CT cells spread out and migrate, forming a simple squamous epithelium: Heuser's membrane, which delimits, with the endophysis, beneath the embryonic disc a cavity called the primary yolk sac (LI) or primitive yolk sac replacing the blastocoel.

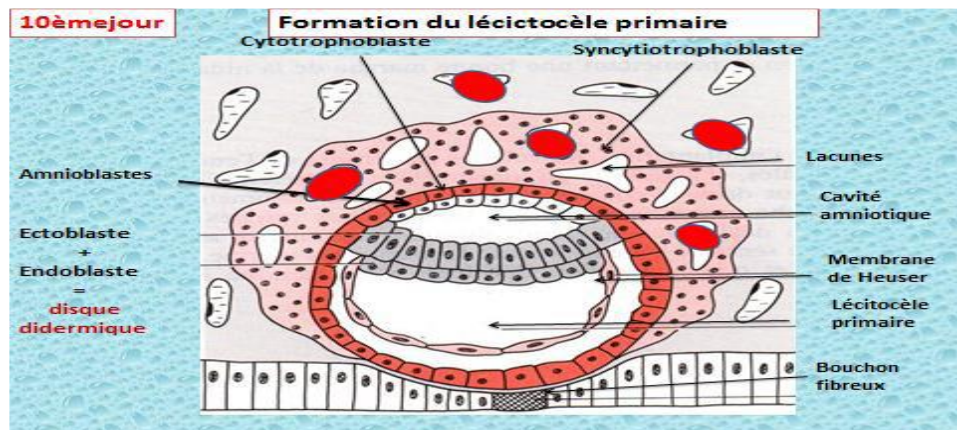


Figure 5. Formation of the primary yolk sac

➤ Formation of extra-embryonic mesenchyme (MEE)

The extra-embryonic mesenchyme is originating from the CT, this very loose tissue filled with cavities insinuates itself between the CT and the amnioblasts. The small fluid cavities within the mesenchyme converge into a single cavity: the chorionic cavity or extra-embryonic coelom.

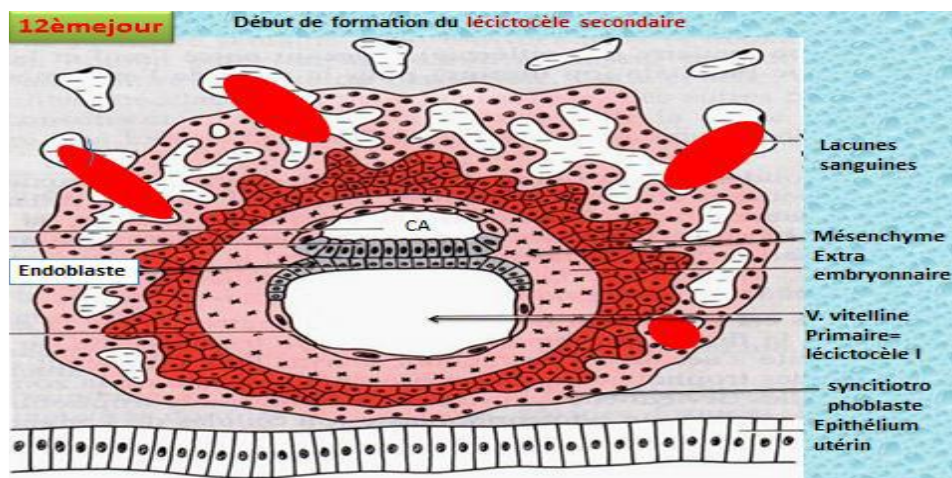


Figure 6. Formation of extra-embryonic mesenchyme

The MEE will form mesenchymal tissue arranged in 4 positions:

- A layer on the inner surface of the CT forming with it, the chorionic plate.
- A layer on the outer surface of the secondary yolk sac (entophyll) forming with it, the extra-embryonic splanchnopleur or umbilical plate.
- A layer on the outer surface of the amniotic cavity (amnion) forming with it, the extra-embryonic somatopleur or amniotic plate.

- A cell mass, between the CT and the amniotic cavity, the embryonic stalk or fixation stalk which connects the trophoblast to the bilaminar disc.

➤ Secondary lecithocele formation

During coelom growth, a new wave of cells, originating from the endoderm, pushes back Heuser's membrane. This proliferation causes a constriction of the primary yolk sac, which divides into 2 cavities: the secondary yolk sac (LII) (or vitelline vesicle), which is entirely lined by endoderm cells, and an exocoelomic cyst, whose growth stops.

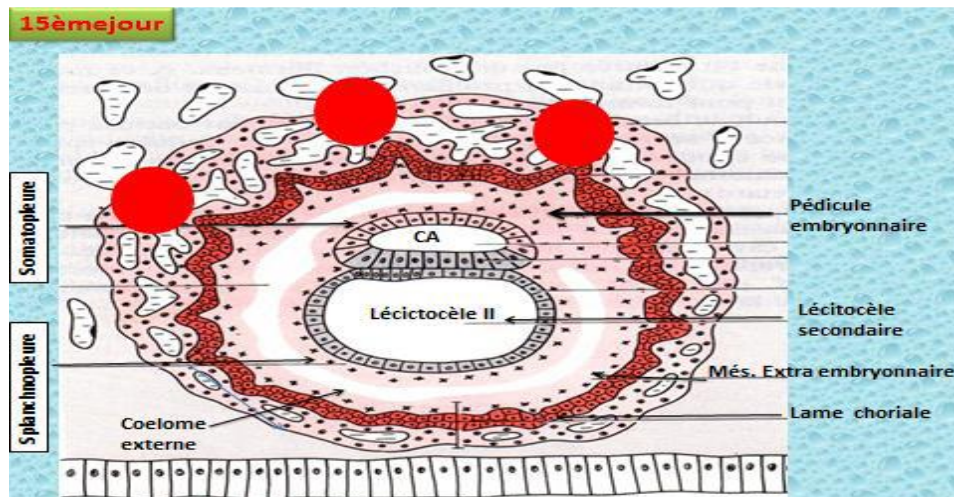


Figure 7. Formation of the secondary yolk sac

5. AT THE END OF THE 2ND WEEK

The egg has a complex structure with:

- In the center, two hollow hemispheres, which are connected to the inner face of the chorionic sphere by the embryonic stalk. An oval, bilaminar embryonic disc, with a diameter of 200µm, composed of the superficial ectophyll and the underlying entophyll.
- the amniotic cavity, above the embryonic disc, whose roof is the amnion and the floor is the ectophyll.
- secondary lecithocele (yolk sac) below, delimited by the cells of the entophyll.
- The whole is enveloped by extra-embryonic mesenchyme, forming the extra-embryonic somatopleur and splanchnopleur.
- This whole is bathed in the coelom, the inner wall of which is formed by the chorionic plate (mesenchyme and cytotrophoblast), which together with the syncytiotrophoblast form the chorion.